



MEV Shanghai

2024

OCT 16TH / 14:00

DEX Arbitrage

The Most Misunderstood
Concept in MEV



DR. MARK
RICHARDSON

Bancor / Carbon DeFi





Bancor



CARBON^{DEFI}

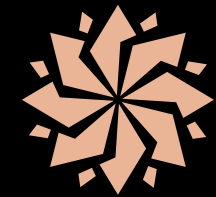
DEX ARBITRAGE

Mark B. Richardson, Ph.D.

Project Lead, Bancor

4:00pm, 16th October 2024

L1-10, Shuangshihui, No. 399, Dongchangzhi Road, Hongkou District, Shanghai



PANTA
RHEI



ecosystem
support
program

Arbitrage 101:

1. Sell token_a for token_b on market_0.
2. Buy token_a with token_b on market_1.
3. token_a sold < token_a bought.

Pool State:

$\{\text{token_0} : x_p, \text{token_1} : y_p, \text{fee} : \delta_p\}$



<Notation>

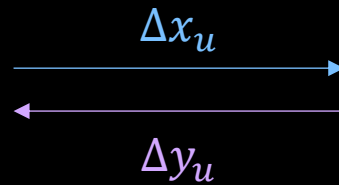
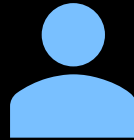
Pool State:

$\{\text{token}_0 : x_p, \text{token}_1 : y_p, \text{fee} : \delta_p\}$



User Wallet State:

$\{\text{token_0} : x_u, \text{token_1} : y_u\}$



Pool State:

$\{\text{token_0} : x_p, \text{token_1} : y_p, \text{fee} : \delta_p\}$

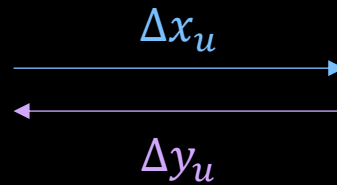
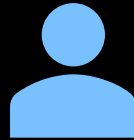


User Wallet State:

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Expected Transaction:

User Wallet: $\{\text{token_0} : x_u - \Delta x_u, \text{token_1} : y_u + \Delta y_u\}$

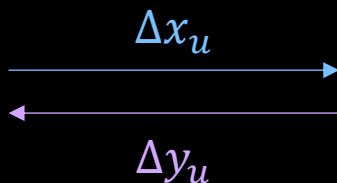
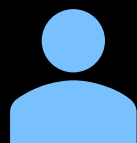
Pool: $\{\text{token_0} : x_p + \Delta x_u, \text{token_1} : y_p - \Delta y_u\}$

User Wallet State:

$\{\text{token_0} : x_u, \text{token_1} : y_u\}$

Pool State:

$\{\text{token_0} : x_p, \text{token_1} : y_p, \text{fee} : \delta_p\}$



So:

$x_u, y_u, \Delta x_u, \Delta y_u, x_p, y_p \in \mathbb{R}^+$,
and $\Delta x_u \leq x_u$, and $\Delta y_u \leq y_p$,
(and, also $0 \leq \delta_p < 1$)

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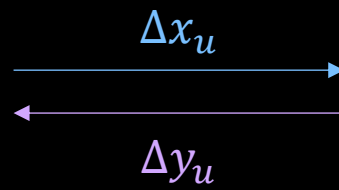
Δx_u
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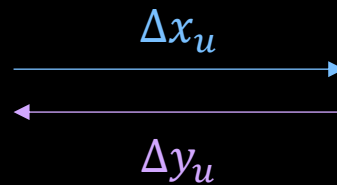
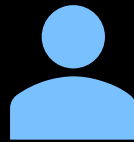


“Feeless” Model:

$$x_p \cdot y_p = k$$

Calculate “Amount Out” from “Amount In”

$$f(\Delta x_u) = \frac{\Delta x_u \cdot y_p}{x_p + \Delta x_u} = \Delta y_u$$



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Calculate “Amount In” from “Amount Out”

$$f(\Delta y_u) = \frac{x_p \cdot \Delta y_u}{y_p - \Delta y_p} = \Delta x_u$$

“Fee from Amount In”:

“Feeless” Model:

Calculate “Amount Out” from “Amount In”

$$f(\Delta x_u) = \frac{\Delta x_u \cdot (1 - \delta_p) \cdot y_p}{x_p + \Delta x_u \cdot (1 - \delta_p)} = \Delta y_u$$

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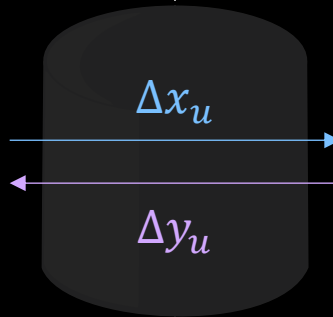
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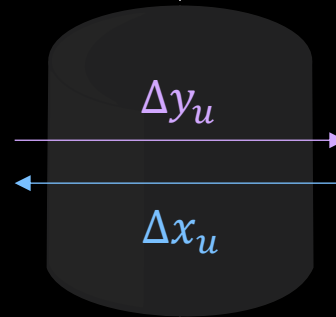
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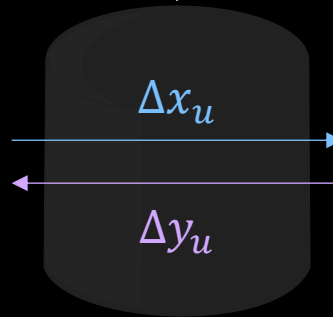
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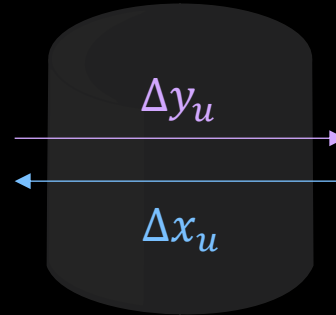
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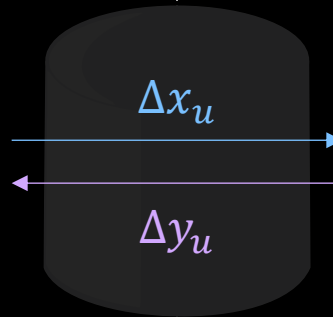
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<\Exchange Model>

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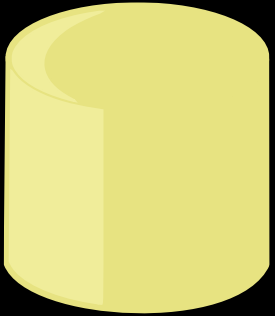
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<Performing DEX Arbitrage>

Fee Model: “Fee from Amount In”



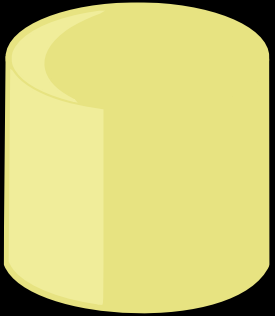
```
pool_0:{  
  x_0 : 123,  
  y_0 : 456,  
  fee : 0.003}
```

Fee Model: “Fee from Amount In”



```
pool_1:{  
  x_1 : 321,  
  y_1 : 765,  
  fee : 0.002}
```

Fee Model: “Fee from Amount In”



```
pool_0:{  
  x_0 : 123,  
  y_0 : 456,  
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$$\frac{\Delta x_0 \cdot y_0}{\frac{x_0}{1 - \delta_0} + \Delta x_0} = \Delta y_0$$

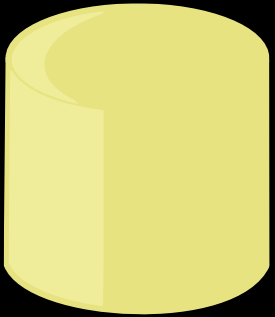
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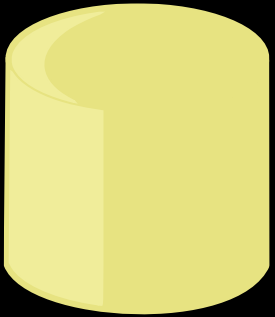
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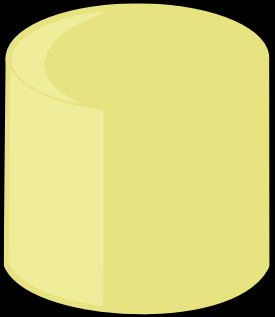
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$$\Delta x_1 - \Delta x_0 = \frac{\frac{\frac{\Delta x_0 \cdot y_0}{x_0} + \Delta x_0}{1 - \delta_0} \cdot x_1}{\frac{y_1}{1 - \delta_1} + \frac{\frac{\Delta x_0 \cdot y_0}{x_0} + \Delta x_0}{1 - \delta_0}} - \Delta x_0$$

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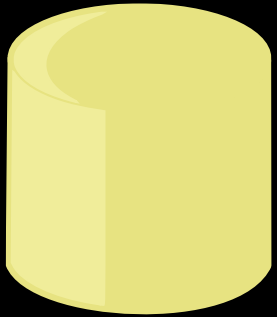


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```

$$\Delta x_1 - \Delta x_0 = \frac{\Delta x_0 \cdot x_1 \cdot y_0}{\left(\Delta x_0 + \frac{x_0}{1 - \delta_0}\right) \cdot \left(\frac{\Delta x_0 \cdot y_0}{\Delta x_0 + \frac{x_0}{1 - \delta_0}} + \frac{y_1}{1 - \delta_1}\right)} - \Delta x_0$$

$$\Delta x_0 \Big|_{\frac{\partial(\Delta x_1 - \Delta x_0)}{\partial \Delta x_0} = 0} = \frac{\delta \cdot x_0 \cdot y_1 - x_0 \cdot y_1 + \sqrt{x_0 \cdot x_1 \cdot y_0 \cdot y_1 \cdot (\delta_0^3 \cdot \delta_1 - \delta_0^3 - 3 \cdot \delta_0^2 \cdot \delta_1 + 3 \cdot \delta_0^2 + 3 \cdot \delta_0 \cdot \delta_1 - 3 \cdot \delta_0 - \delta_1 + 1)}}{-\delta_0^2 \cdot \delta_1 \cdot y_0 + \delta_0^2 \cdot y_0 + \delta_0^2 \cdot y_1 + 2 \cdot \delta_0 \cdot \delta_1 \cdot y_0 - 2 \cdot \delta_0 \cdot y_0 - 2 \cdot \delta_0 \cdot y_1 - \delta_1 \cdot y_0 + y_0 + y_1}$$

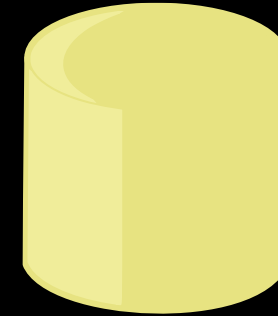
Fee Model: “Fee from Amount In”



pool_0:{
x_0 : 123,
y_0 : 456,
fee : 0.003}

$$\Delta x_0 = 18.884044775539138$$

$$\Delta y_0 = 60.533377117857$$



pool_0:{
x_0 : 141.8840,
y_0 : 395.4666,
fee : 0.003}

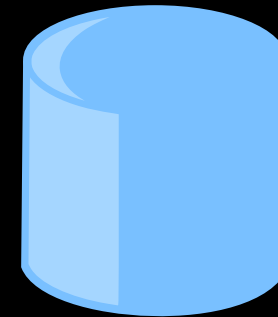
Fee Model: “Fee from Amount In”



pool_1:{
x_1 : 321,
y_1 : 765,
fee : 0.002}

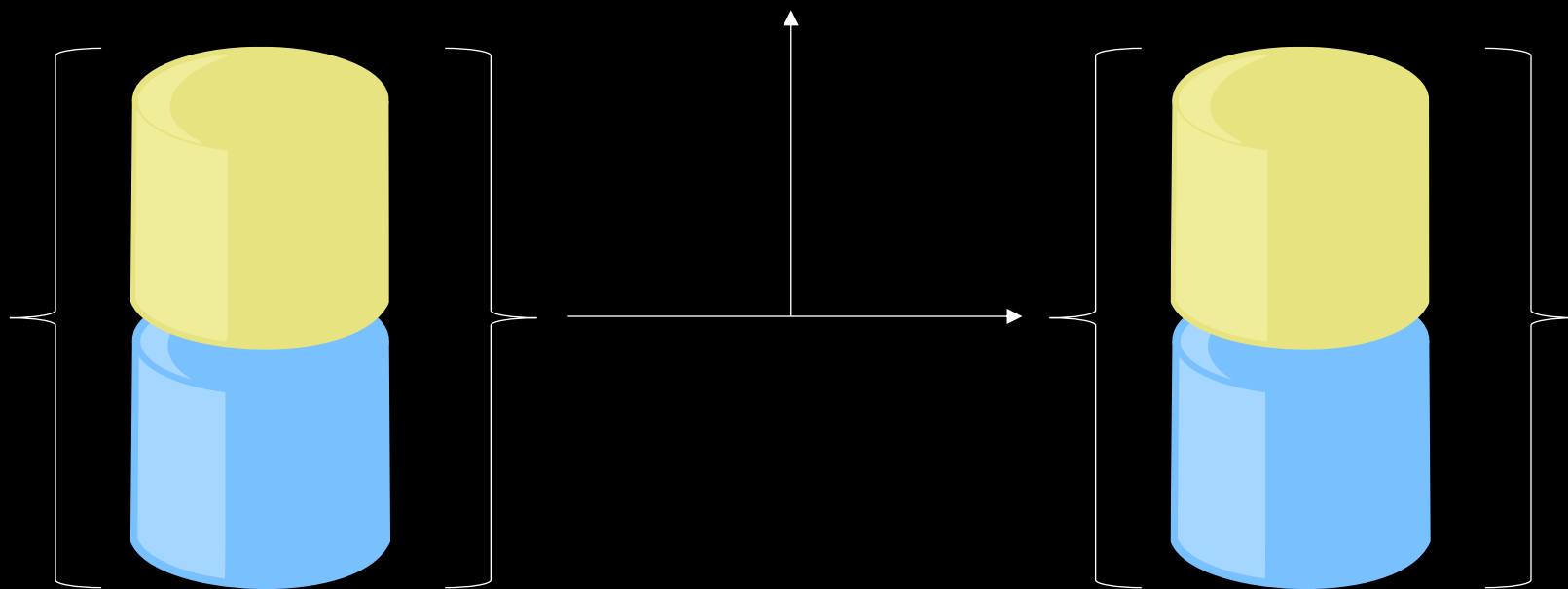
$$\Delta y_1 = 60.533377117857$$

$$\Delta x_1 = 23.4941390905346$$



pool_1:{
x_1 : 297.5059,
y_1 : 825.5334,
fee : 0.002}

$$\Delta x_1 - \Delta x_0 = 4.6100943149955$$



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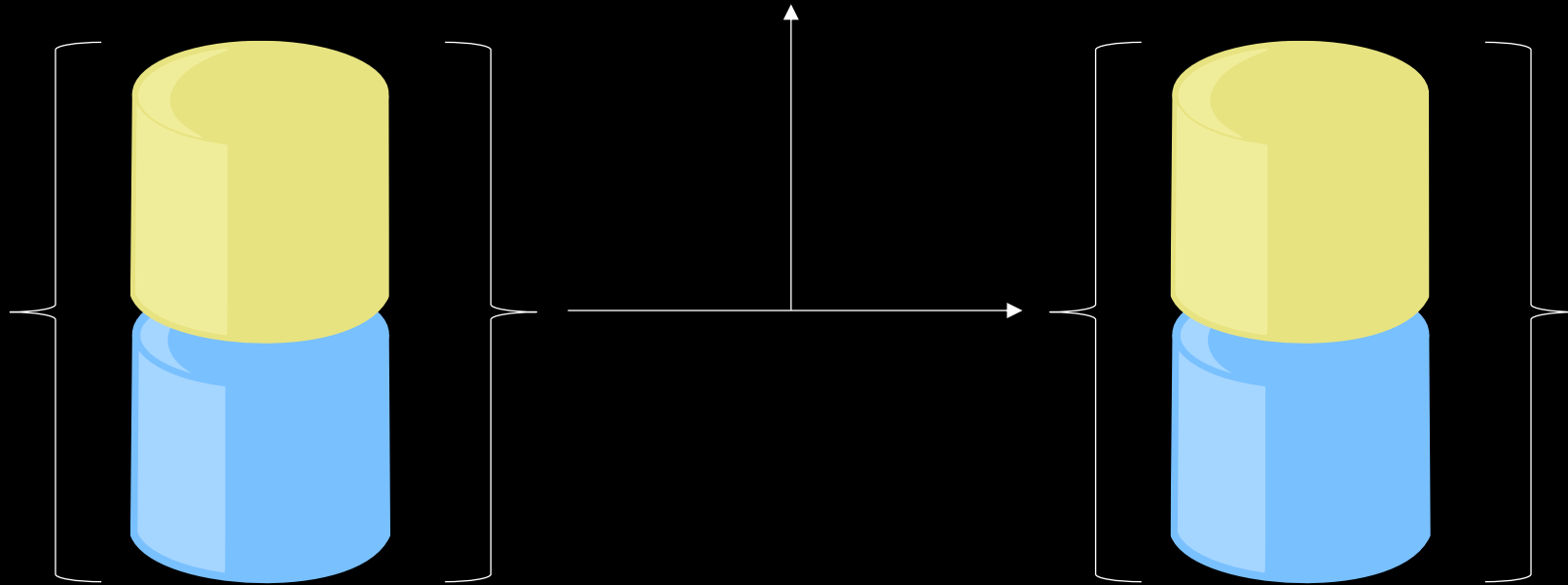


</Performing DEX Arbitrage>

$$\Delta x_1 - \Delta x_0 = 4.6100943149955$$

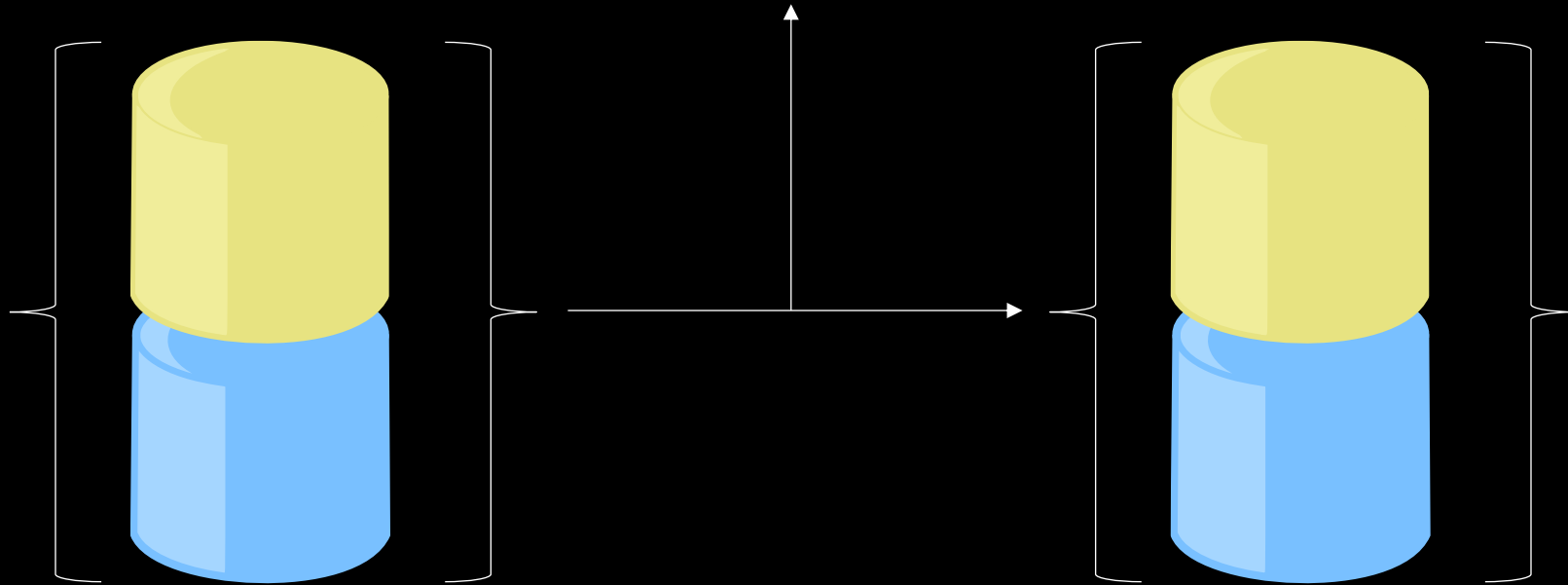


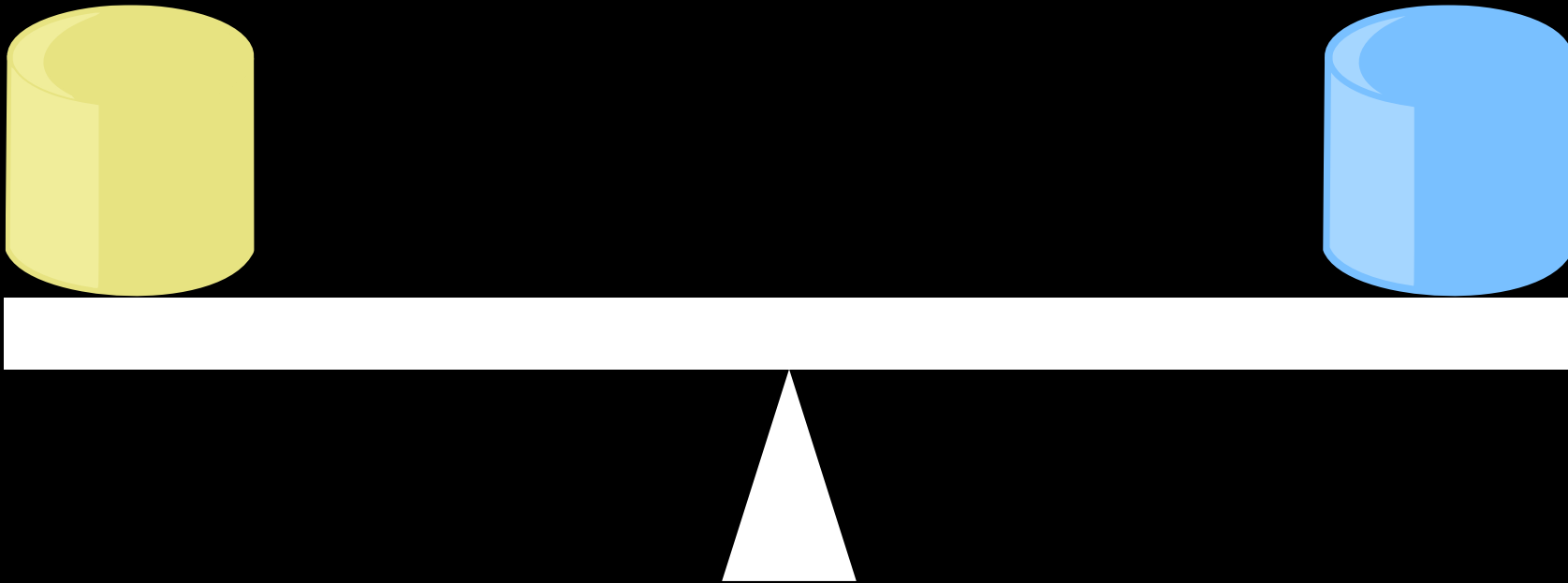
“Arbitrage trading extracts value from liquidity providers”

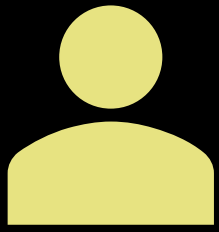


“Arbitrage trading extracts value from liquidity providers”

It's not that simple...

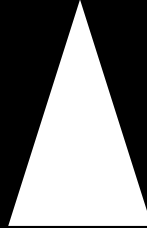
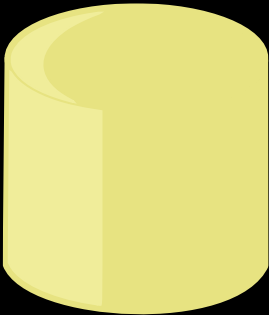




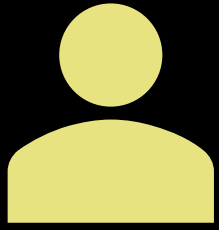


interacts via DEX front-end

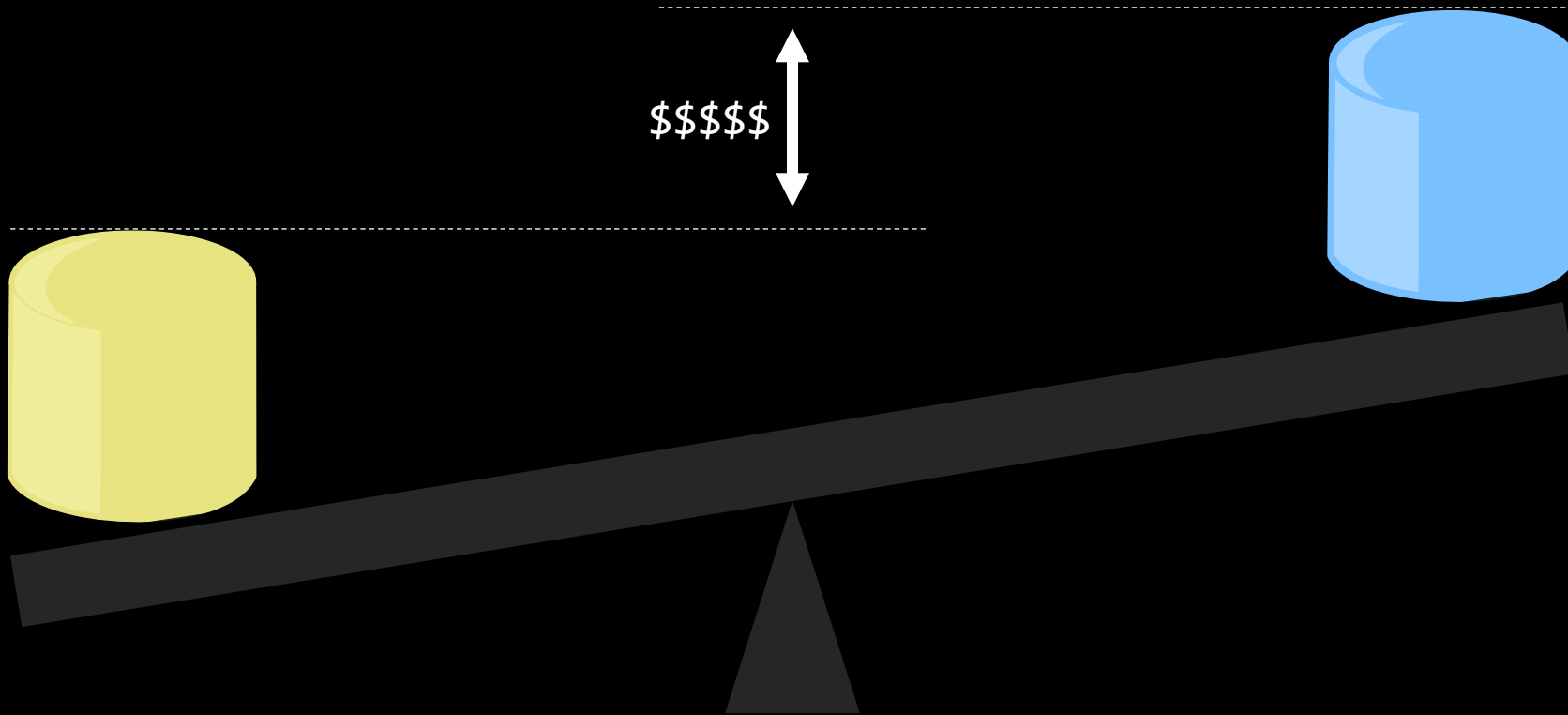
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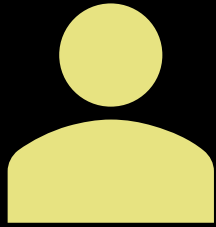




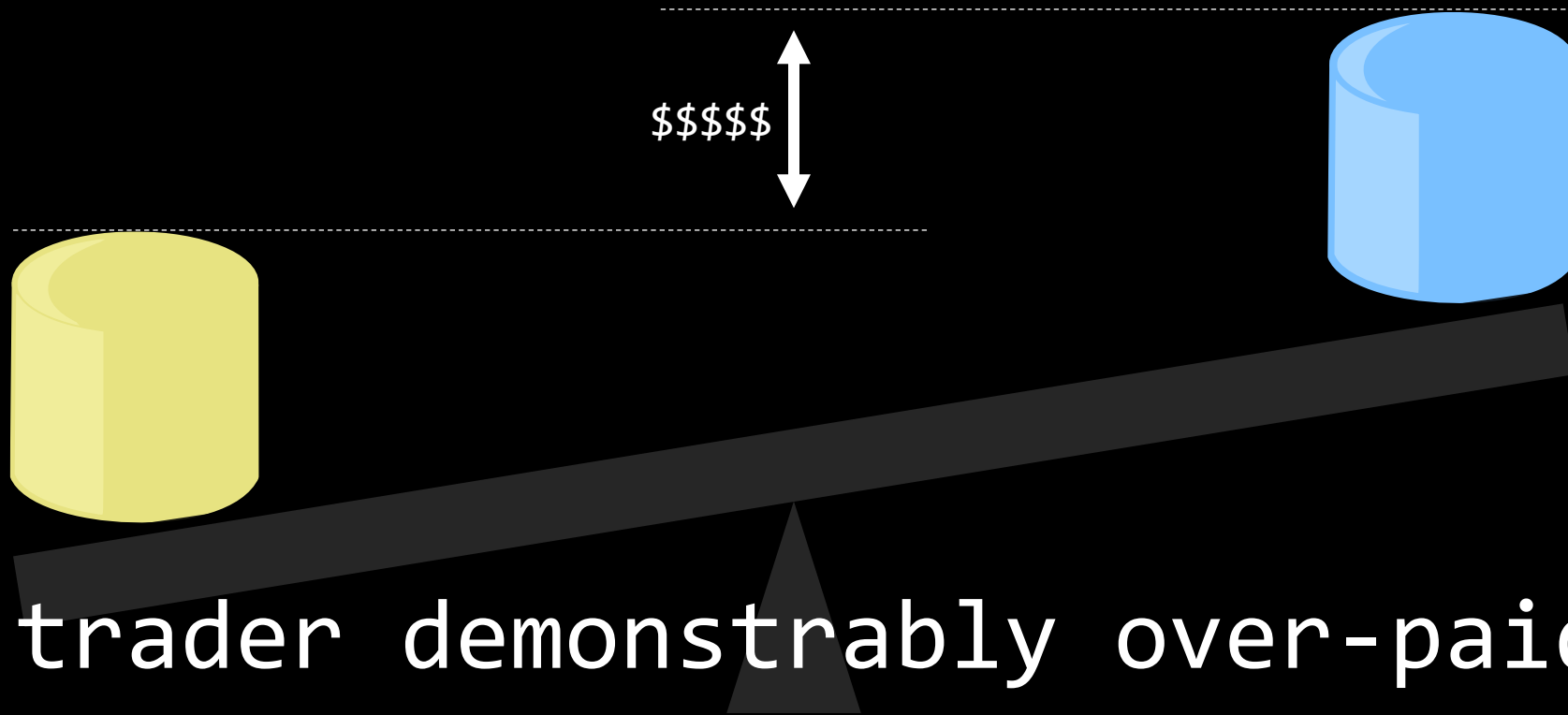


This discrepancy was paid
by the trader.

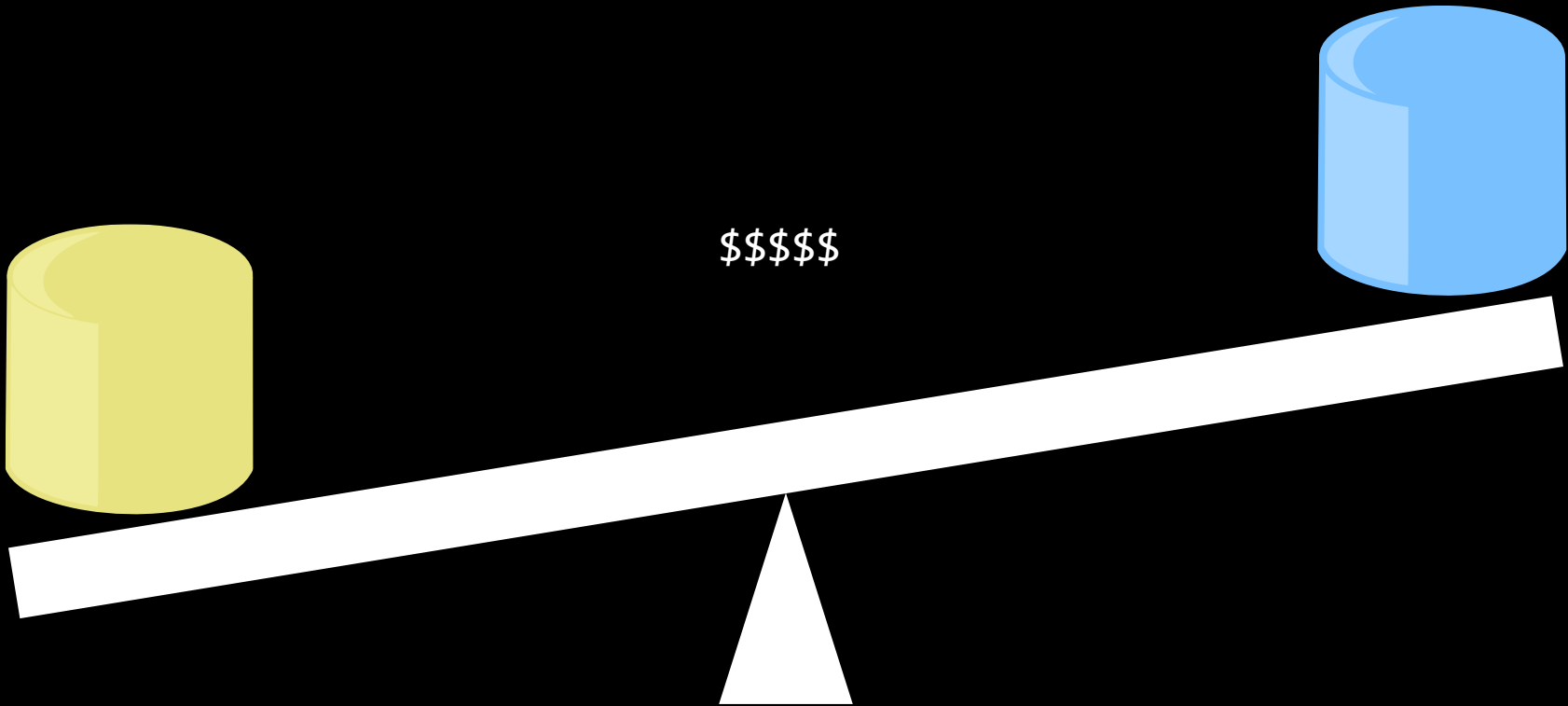
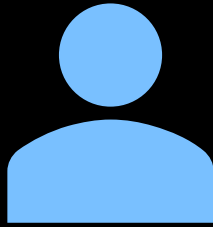
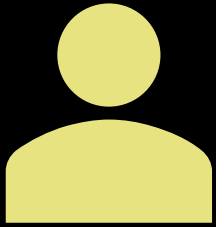


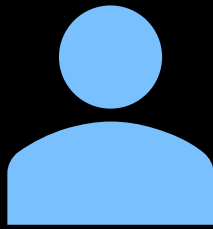
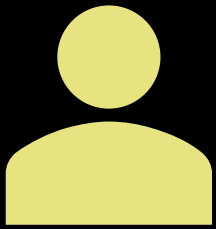


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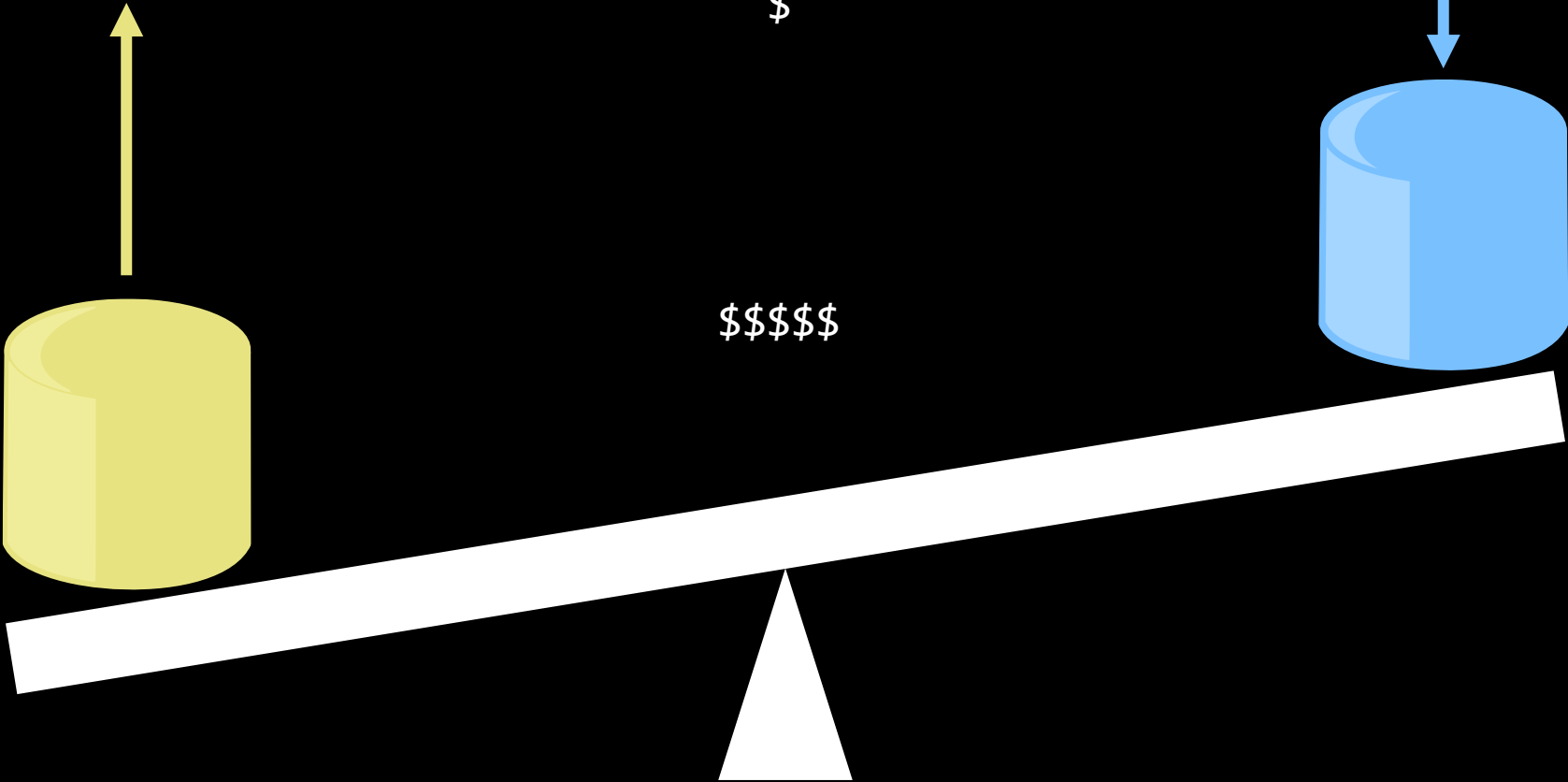
The trader demonstrably over-paid for
their tokens, relative to the available
market.

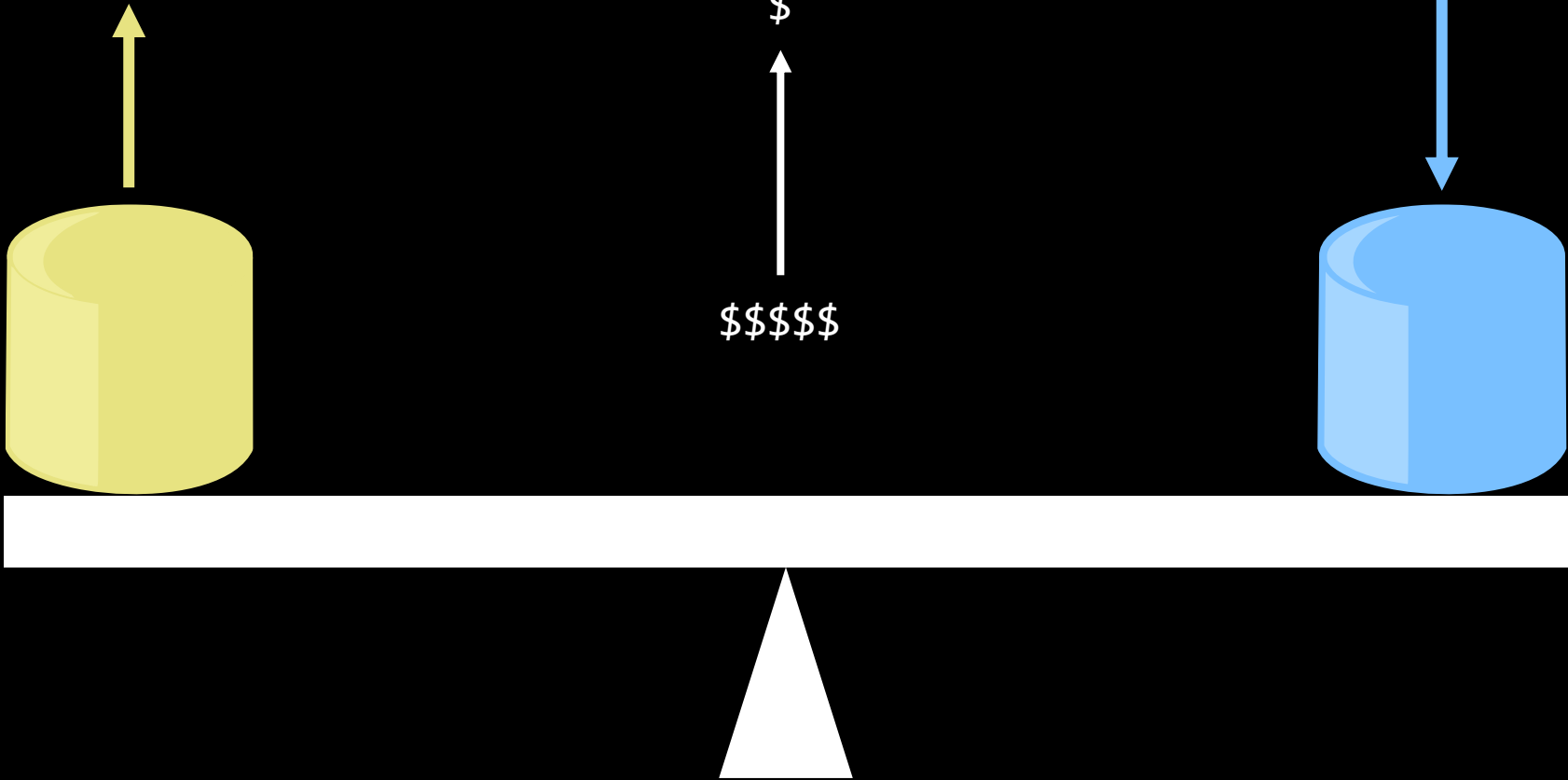
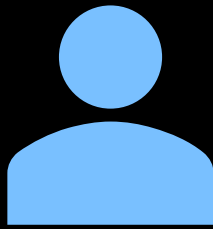
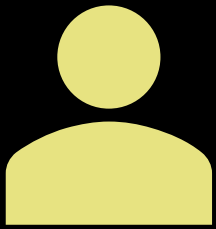


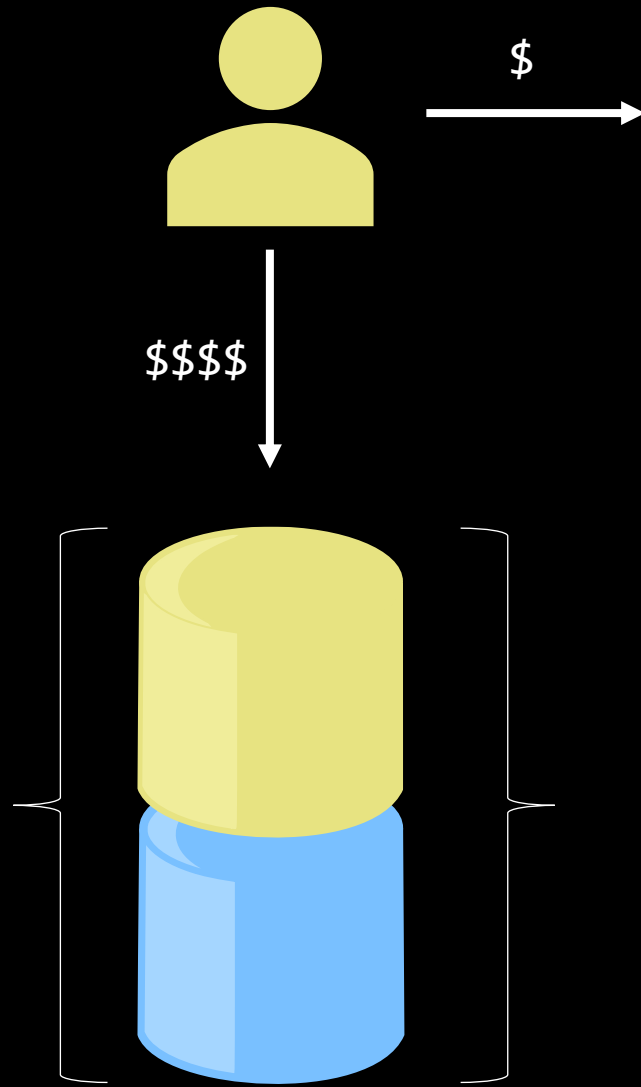


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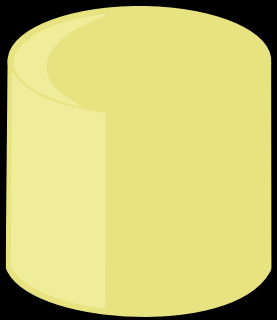


The over-paid amount is
[almost] equal to the
arbitrage value.

Liquidity providers are
not a victim here.

On outcome indifference:

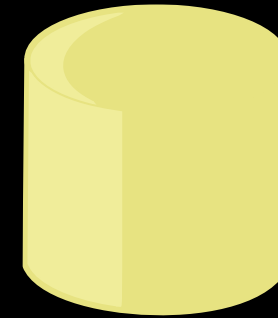
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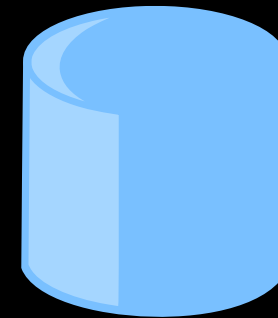
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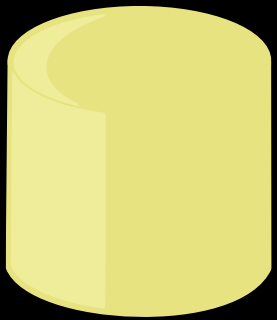
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Assume these two trades were performed by different people.

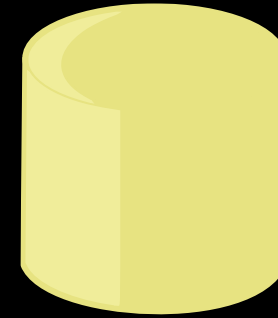
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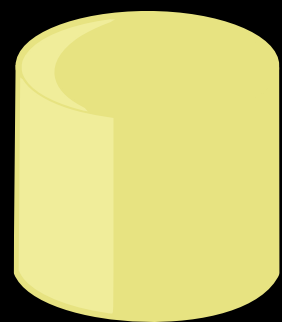
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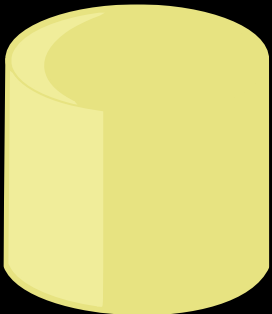
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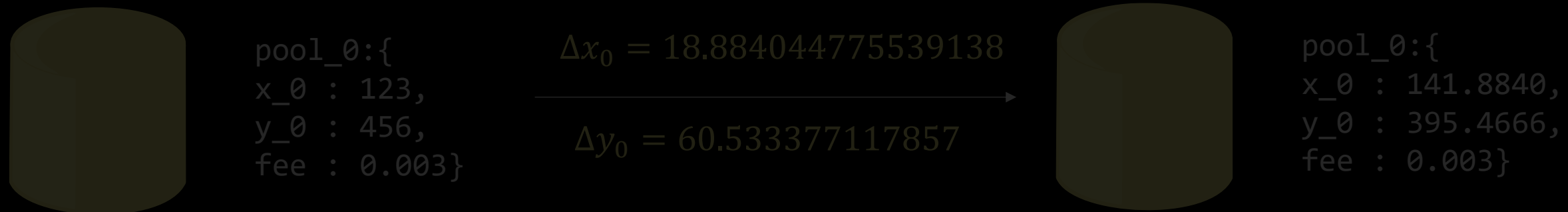
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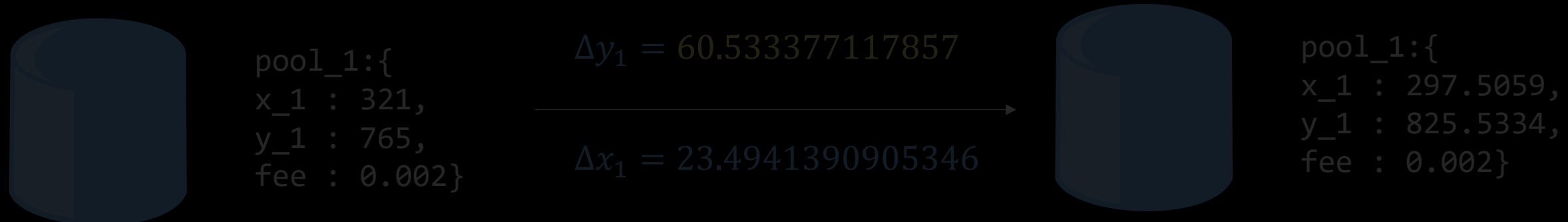
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If it doesn't change the outcome, then it doesn't matter.

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<The Misconception>

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MB_Richardson



MBRichardson87



mrichardson87



mark@bancor.network



MEV Shanghai

2024

OCT 16TH / 14:00

DEX Arbitrage

The Most Misunderstood Concept in MEV



DR. MARK
RICHARDSON
Bancor / Carbon DeFi

